There are primarily two types of objects in IoT communication — publisher and subscriber. Publishing refers to the transmission of data signals — the process of moving data from a computer to an external device. The receiver system is referred to as the subscriber in IoT communication.

Publisher and subscriber are terms used to describe the source and recipient of a data signal, respectively. The MQTT broker, also known as an IoT server, serves as the focal point and regulates communication between publishers and subscribers. It manages the data that publishers broadcast for the other devices known as subscribers. For instance, information may be sent from a temperature sensor to a farmer's mobile phone. In this instance, the temperature sensor (publisher) is the source of the data signals, and the farmer's gadget is the receiver (subscriber).

These MQTT broker applications can be installed on a variety of specialised hardware, including Raspberry Pi, Arduino, laptops, servers, and many others.

Cloud based MQTT

URL: cloudmqtt.com

CloudMQTT, one of the most widely used MQTT brokers, can be used for device communication and implementation of Industry 4.0 based scenarios. It can be used to implement IoT scenarios for a variety of applications including smart toll plaza, smart cities, smart homes, and smart offices.

For researchers and academicians, CloudMQTT offers a free package called 'Cute Cat' that allows five users or connections for IoT devices. Using an active Google account, free instances can be built on the CloudMQTT network.

The features integrated in Cloud MQTT include:

- Open source message broker that implements the MQTT protocol
- 24x7 availability of MQTT broker
- Cloud compatibility
- Lightweight and high performance; integrates with low power single boards like Raspberry Pi, Arduino, and ESP8266 as well as computers and servers
- Integration of lightweight methods for messaging using a publish and subscribe message queueing model
- Fully managed Mosquitto servers in the cloud

To establish a real-time connection between CloudMQTT and a device, IoT hardware or smartphones can use authentication information from CloudMQTT.

Traffic simulation using Eclipse SUMO

URL: eclipse.org/sumo

Smart vehicular communication in smart cities and e-governance are key application areas of Industry 4.0. Eclipse SUMO (Simulation of Urban MObility) is very effective in simulating a traffic scenario. It integrates a range of features to simulate smart vehicular environment, and can work with multiple vehicles of different types in different environmental conditions.

SUMO is free and open source. Since its introduction in 2001, it has enabled the modelling of multi-mode traffic networks, which include pedestrians, public transportation, and private vehicles. Numerous auxiliary tools, such as network import, route calculations, visualisation, and emission

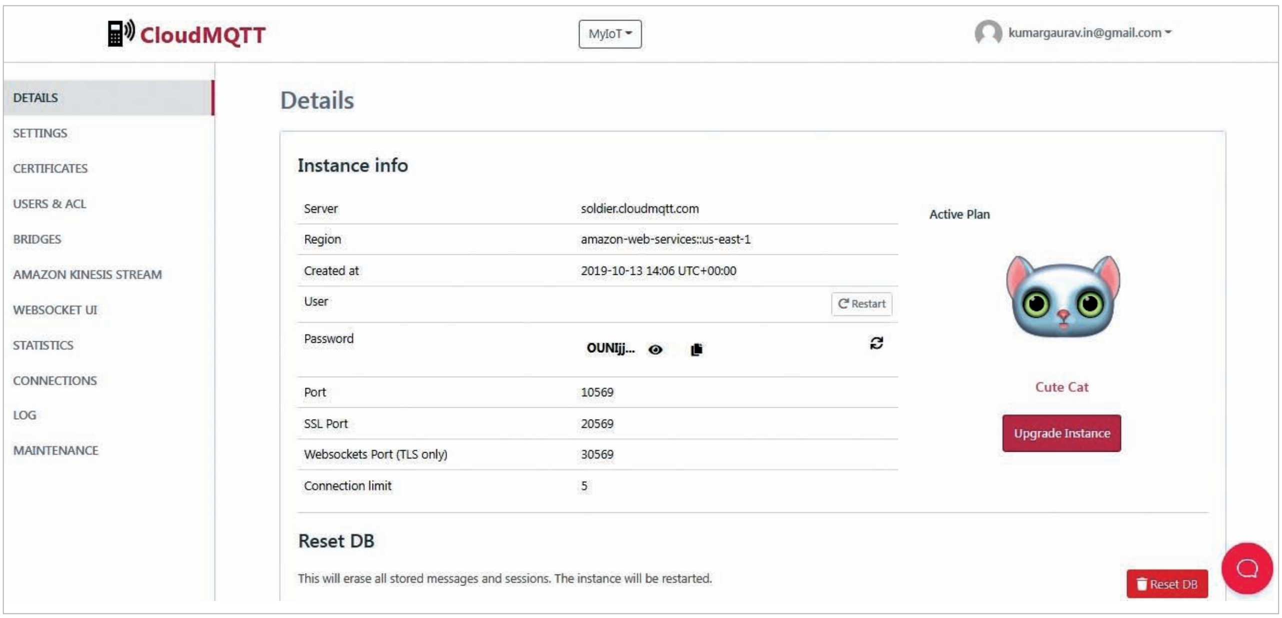

Figure 2: Cloud based MQTT for simulating an IoT scenario